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1. PROJECT TITLE

AI-Driven Civic Issue Reporting and Management System with Geotagging and Worker Tracking

2. ABSTRACT

The rapid urbanization of cities has led to increased challenges in managing civic infrastructure. Citizens frequently encounter issues such as potholes, broken streetlights, uncollected garbage, and water supply disruptions, but traditional reporting mechanisms are often slow, opaque, and inefficient. This project proposes an AI-Driven Civic Issue Reporting and Management System to streamline this process. The system allows citizens to capture and upload images of civic issues using a geotag-enabled camera, which automatically extracts precise location coordinates. A custom-trained Image Recognition model (eliminating dependency on external APIs like Gemini AI) automatically categorizes the issue and routes the complaint to specialized municipal departments (e.g., Water, Garbage, Streetlight, Pothole). To ensure accountability, the system features full tracking of complaint resolution progress displayed as a percentage. Additionally, it incorporates a comprehensive municipal worker registration and tracking module, allowing authorities to assign tasks and monitor worker locations and progress in real-time. The expected outcome is a transparent, efficient, and automated e-governance platform that reduces resolution times, optimizes resource allocation, and enhances citizen satisfaction.

3. INTRODUCTION / BACKGROUND

Effective civic infrastructure management is a cornerstone of smart city development. As urban populations grow, municipal corporations face mounting pressure to maintain public amenities and address citizen grievances promptly. Traditionally, reporting civic issues involves manual data entry, physical visits, or inefficient helplines, leading to misclassification of complaints and delayed resolutions. The advent of artificial intelligence, specifically computer vision, and ubiquitous smartphone adoption presents a unique opportunity to modernize e-governance. By integrating automated image recognition, precise geotagging, and real-time worker tracking, municipalities can transition from reactive to proactive maintenance. This project is situated in the domain of Smart City Governance, aiming to bridge the communication gap between citizens and local authorities through a robust, automated, and highly accountable digital platform.

4. PROBLEM STATEMENT

The lack of an efficient, transparent, and automated system for reporting, categorizing, and tracking civic issues leads to delayed resolutions, misallocation of municipal resources, and reduced citizen satisfaction.

5. OBJECTIVES

O1: To develop a citizen-facing application featuring a geotag-enabled camera that automatically captures and attaches precise location data to reported issues.
O2: To train and integrate a custom Image Recognition model capable of automatically identifying and categorizing civic issues (e.g., potholes, garbage, broken streetlights) to route them to the respective municipal departments.

O3: To implement a transparent complaint tracking mechanism that provides real-time updates on the resolution progress in percentage terms.
O4: To create a dedicated worker management module for the registration, task assignment, and real-time tracking of municipal workers to ensure operational accountability.

6. MOTIVATION AND NEED

Currently, municipal grievance redressal systems suffer from high latency and human error in complaint categorization. Citizens often feel disconnected from the process due to a lack of visibility into the status of their complaints. Furthermore, municipal authorities struggle to monitor their field workers effectively, leading to inefficiencies in task execution. There is a pressing need for a system that not only makes reporting effortless for the user—by automating location fetching and issue classification—but also enforces accountability on the administration side. By building a custom AI model rather than relying on third-party APIs, the system will ensure data privacy, reduce recurring operational costs, and offer tailored recognition for local infrastructural issues. This system is highly relevant technologically and socially, aligning with the broader vision of smart, responsive cities.

7. EXISTING SOLUTIONS / BRIEF LITERATURE SURVEY

The integration of Information and Communication Technology (ICT) into civic management has been extensively researched in recent years. Traditional e-governance portals often rely on text-based descriptions, which are prone to ambiguity, misinterpretation, and delayed routing. Recent studies emphasize the paradigm shift towards crowdsensing, automated issue classification, and transparent municipal tracking.

Sharma and Kumar (1) proposed a deep learning-based pothole detection system utilizing Convolutional Neural Networks (CNNs) for smart cities. Their research demonstrated high accuracy in identifying road anomalies, yet it lacked an integrated workflow for municipal worker assignment and citizen feedback. Similarly, Chen et al. (3) explored automated solid waste classification using image processing. They highlighted the efficiency of custom-trained computer vision models over manual sorting and tagging. However, their scope was strictly limited to waste management rather than offering a holistic civic portal covering multiple utility domains.

Gupta and Singh (2) reviewed various e-governance frameworks across developing nations, noting that the absence of real-time tracking is a primary cause of citizen dissatisfaction and institutional distrust. To address field operations and personnel management, Patel and Desai (4) developed an IoT and GPS-based framework for tracking municipal workers. Their empirical findings proved that real-time monitoring significantly boosts task completion rates and curtails resource mismanagement. Nonetheless, their system operated independently and was not coupled directly with an automated AI-driven complaint routing mechanism originating from citizen inputs.

Furthermore, Kim and Lee (8) explored the intelligent routing of complaints using image analysis and natural language processing. While effective, their reliance on generalized, paid cloud AI services (analogous to Gemini or OpenAI) raised concerns about ongoing operational costs, continuous internet dependency, and potential data privacy breaches. In a parallel study on spatial accuracy, Kumar et al. (9) emphasized the critical role of precise geotagging in urban crowdsourcing. They noted that manual location entry by citizens often results in unresolvable complaints due to spatial inaccuracies and ambiguous landmark references.

Additionally, Zhang et al. (10) proposed an IoT-based framework for streetlight management, showing how isolated departmental databases can improve response times. Al-Hader and Rodzi (5) also discussed the importance of decentralized departmental data processing in smart city infrastructure development.

This proposed project synthesizes these diverse approaches by combining automated, precise geotagging (9), custom image recognition covering multiple distinct categories (1, 3, 6), and comprehensive real-time worker tracking (4). Unlike existing solutions that address these components in isolation or rely heavily on third-party API solutions (8), the proposed system offers a unified, autonomous, and cost-effective platform. By establishing separate departmental queues (Water, Garbage, Streetlight, Potholes) and implementing percentage-based progress tracking, the project directly addresses the operational gaps identified in the current literature.

Sr. No.	Source / Existing Solution	Approach	Key Features	Limitations
1	Sharma & Kumar (2021)	CNN for pothole detection	High accuracy image classification	Limited to potholes; no worker tracking
2	Patel & Desai (2023)	GPS tracking for workers	Real-time field tracking	Not integrated with citizen complaints
3	Kim & Lee (2021)	NLP & Image Analysis API	Automated routing	Relies on paid third-party AI APIs
4	Existing City Portals	Manual web forms	Centralized data collection	Lacks AI categorization & live tracking

8. PROPOSED SOLUTION

The proposed solution is a comprehensive AI-Driven Civic Issue Management platform comprising three main interfaces: a Citizen App, a Departmental Dashboard, and a Worker App.
When a citizen encounters a civic issue, they use the application's built-in camera to capture an image. The system automatically fetches the precise GPS coordinates via a geotagging module. A custom-trained machine learning model processes the image to identify the issue (e.g., Garbage, Water Leak, Pothole) and automatically forwards the complaint to the newly separated, specialized municipal departments (Water Department, Garbage Department, Broken Streetlight Department, Pothole Department). Once routed, department admins can assign the task to registered workers. The citizen can track the live progress of their complaint, displayed as a completion percentage (e.g., 25% Assigned, 50% In Progress, 100% Resolved). Concurrently, the system tracks the registered workers' locations and task statuses, ensuring full transparency and operational accountability without relying on external LLMs like Gemini AI for issue classification.

9. PROPOSED METHODOLOGY / APPROACH

1. **Problem Identification & Requirements Gathering**: Defining issue categories and municipal workflow requirements.
2. **Data Collection & Model Training**: Gathering a diverse dataset of civic issues (images of potholes, garbage, etc.) and training a custom Convolutional Neural Network (CNN) for image recognition.
3. **System Design**: Designing the architecture, database schema, and separate department logic.
4. **Development**: Building the citizen interface with geotagging, the worker tracking module, and the departmental dashboards. Integrating the trained AI model.
5. **Testing**: Conducting functional, geolocation accuracy, and AI model precision testing.
6. **Evaluation & Refinement**: Analyzing performance metrics and user feedback to refine the system.

10. PROPOSED TECHNOLOGY / TOOLS

Component	Proposed Technology / Tool	Justification
Frontend	React.js / Next.js	Enables fast, responsive, and dynamic user interfaces for both citizens and admins.
Backend	Node.js with Express.js	Scalable and non-blocking architecture suitable for handling multiple concurrent complaints.
Database	MongoDB	Flexible schema design ideal for storing varying complaint data, geotags, and user profiles.
AI Model	Python, TensorFlow / PyTorch	Industry standard for training custom robust image classification models.
Geolocation	HTML5 Geolocation API / Map API	Highly accurate location fetching and visual mapping of complaints and workers.

11. EXPECTED OUTCOMES / DELIVERABLES

Deliverable	Description
Citizen Application	Web/Mobile interface with geotag camera and percentage-based complaint tracking.
Custom AI Model	A trained image recognition model capable of categorizing civic issues autonomously.
Departmental Dashboard	Admin panel for routing, managing, and overseeing complaints per department.
Worker Management Module	System for registering workers, assigning tasks, and tracking their locations.

12. PRELIMINARY EVALUATION PLAN

The system will be evaluated across three dimensions:

- **AI Model Performance**: Measured using standard ML metrics including Accuracy, Precision, Recall, and F1-Score on a holdout test dataset.
- **Functional Testing**: Verifying that the geotagging module accurately captures coordinates within a 5-meter radius and that complaints route to the correct isolated department.
- **Usability & Workflow**: Conducting user testing to ensure citizens can report issues in under 3 steps and verifying that the percentage tracking updates accurately based on worker input.

13. FEASIBILITY ANALYSIS

- **Technical Feasibility**: High. The proposed stack (MERN + Python ML) is well-documented and widely supported.
- **Resource Feasibility**: High. Open-source datasets for urban issues are available to bootstrap the custom ML model.
- **Skill Feasibility**: The project aligns with standard software engineering and applied machine learning skill sets.
- **Time Feasibility**: Achievable within a standard academic semester provided a strict adherence to the proposed timeline.
- **Risk**: The primary risk is achieving high accuracy in the custom image recognition model across varying lighting conditions. This will be mitigated by implementing data augmentation techniques during training.

14. PROJECT SCOPE

14.1 In Scope

- Geotag-enabled camera integration for automated location tracking.
- Custom image recognition model for categorizing 4-5 specific civic issues.
- Separation of complaints into distinct departmental queues.
- Percentage-based progress tracking for citizens.
- Worker registration and real-time location/status tracking.

14.2 Out of Scope

- Handling emergency services (e.g., Police, Fire, Ambulance).
- Automated physical repair or deployment of autonomous drones.
- Financial transactions, billing, or municipal tax collection.
- Voice-based complaint registration.

15. TENTATIVE TIMELINE

Phase	Activities	Duration
Phase 1	Problem Identification, Data Gathering & Literature Study	Weeks 1–2
Phase 2	Requirements & High-Level Architecture Design	Weeks 3–4
Phase 3	AI Model Training & Backend Development	Weeks 5–10
Phase 4	Frontend Integration, Geotagging & Worker Module	Weeks 11–12
Phase 5	System Testing, AI Evaluation & Refinement	Weeks 13–14
Phase 6	Final Documentation, Report & Presentation	Weeks 15–16

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